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Debt and Money Markets | Semester 2, 2026

ASSESSMENT 3

**Cashflow Matching and Australian Government
Bond Yield-Curve Scenarios**

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Prepared for the Head of Debt

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Executive Summary

This report assesses liability-focused bond construction and the Australian Government Bond outlook through December 2026. Cashflow matching provides a disciplined funding framework when obligations and asset payments are known. Its efficiency depends on settlement costs, eligible instruments, payment timing and trading constraints. A hybrid approach is recommended: secure essential payments first, then manage residual risk within the investment mandate.

Research available by 4 October supports a cautious interest-rate stance. The cash rate is 4.60%, headline inflation is 4.0%, and underlying inflation is 3.6%. The central scenario assumes another 25-basis-point increase in November followed by a December hold. The projected ten-year yield reaches 5.49%, compared with an official public-data baseline of 5.277% on 21 September. These forecasts are analytical judgements, not observed outcomes. Actual Week 11 trading performance and the required LSEG evidence remain unverified because the group records were not supplied.

1. Introduction

The purpose is to advise the Head of Debt on funding specified liabilities economically while recognising changes in the Australian yield curve. Part 1 evaluates construction, cost and control requirements; Part 2 develops dated macroeconomic scenarios and their portfolio implications. The information cutoff is 4 October 2026. The 21 September curve is a historical comparison, not the forecast information cutoff. RBA observations provide a disclosed research substitute for LSEG. The missing liability workbook and execution records prevent conclusions about the group's realised funding cost, purchases or workshop experience.

2. Part 1: Cashflow Matching Portfolio Management

2.1 Investment objective and portfolio design

For a liability-based mandate, success means making required payments when due at an acceptable initial cost. A high portfolio yield alone does not establish success: a bond can offer attractive income while returning principal after the funding obligation. CFA Institute (2026c) distinguishes cashflow matching and duration matching as the principal approaches to liability-focused construction. The appropriate starting point is therefore the liability schedule, followed by the eligible investment universe and dealing constraints.

The proposed process maps coupon and principal payments to individual liability dates, then selects quantities that prevent a cash deficit. Payments must be available before the obligation falls due; matching annual totals is inadequate where payment dates differ. Surplus cash can bridge later obligations only when the mandate permits it. A conservative base calculation credits no interest on that surplus, avoiding dependence on a future reinvestment rate. Eligibility, currency, settlement timing and any minimum trade size must be established before comparing candidate portfolios.

2.2 Selection, quantities and dealing prices

Bond selection should compare the entire payment stream with the liabilities rather than rank instruments by coupon or nominal price. Principal repayments fund larger obligations, while coupons reduce the quantity needed at earlier dates. Discounted-cashflow valuation links a bond's price to those payments and the required return; yield to maturity is the rate that reconciles price and promised cashflows (CFA Institute 2026a). A higher coupon may reduce funding needs at some dates but create unwanted cash at others. Quantities must reflect these interactions.

Dealing evidence must identify the security, trade date, settlement date, purchased face value and executable quote. Where the displayed price excludes accrued interest, the settlement amount includes it; where a quote already includes accrued interest, adding it again overstates cost. Treasury Bond conventions require particular care around the seven-day ex-interest period, when the buyer does not receive the next coupon, and around near-maturity pricing (AOFM n.d.). Counting an unavailable coupon would produce a false funding surplus. The portfolio calculation must follow the actual instrument's convention.

$$\text{Initial cost} = \Sigma [\text{purchased face value} \times \text{settlement price per } \$100 / 100] + \text{explicit transaction costs}$$

For implementation, compare executable purchase terms for feasible substitutes on the same date. Indicative mid-prices are useful for valuation but cannot prove the cost of an executed purchase. After applying trade-size rounding, recalculate every payment-date balance. A solution that works with fractional holdings may fail once the permitted dealing units are imposed. These checks link the quantity rationale to the transaction evidence required by the briefing.

2.3 Lowest-cost construction and funding controls

A defensible minimum-cost claim requires an objective, constraints and evidence of the solution. Cashflow matching can be formulated as a linear programme that minimises

purchase cost subject to funding obligations and non-negative holdings (Kocherlakota, Rosenbloom and Shiu 1990). For this assignment, the proposed extension tracks surplus cash between payment dates and excludes borrowing unless explicitly permitted. If holdings must be integer trading lots, the executable problem also requires those restrictions; the fractional result is only a lower bound.

$$\text{Minimise } C = \sum p_j x_j + \text{costs, subject to } s_t = s_{t-1} + \sum a_{ij} x_j - L_t \geq 0; x_j \geq 0$$

Here, x is purchased quantity, p is settlement cost per unit, a is the unit cashflow available at a payment date, L is the liability and s is unspent cash. The recurrence assumes zero interest on surplus balances. With a defined opening cash balance, the constraint ensures that future payments never fund earlier obligations. Compare alternatives under identical prices, cash treatment and eligibility conditions. Record the resulting cost, minimum cash balance and excess terminal cash. A cheaper portfolio that breaches a payment-date constraint is infeasible.

The numerical illustration in Appendix A shows how later coupons reduce earlier purchases. It is an independently constructed teaching example, not a reconstructed Week 11 transaction. Its cost of 269.11 funds three hypothetical payments of 100 exactly under fractional-holding assumptions. The result demonstrates the calculation method; it does not establish the group's minimum cost. A proper evaluation must reproduce the same reconciliation using the supplied workshop workbook and actual quotations.

2.4 Limitations and alternative approaches

Matching promised payments reduces reliance on selling assets at uncertain future prices, but several vulnerabilities remain. Credit deterioration can interrupt the expected income stream. Callable or otherwise contingent payments weaken certainty. Liability changes can invalidate an initially accurate schedule, while a concentrated maturity structure can limit implementation flexibility. Even a nominal match will not protect purchasing power if obligations are inflation-linked. The relevant response depends on the mandate: reject uncertain payments where appropriate, stress funding assumptions and maintain a defined liquidity reserve rather than treating all coupons as guaranteed cash.

Duration-based immunisation offers greater security-selection flexibility but requires maintenance as time and yields change. For multiple liabilities, matching asset and liability basis-point values is central, alongside adequate asset value and appropriate cashflow dispersion and convexity (CFA Institute 2026b). A duration match should therefore not be described as an exact payment match. A simple ladder also supplies recurring maturities, but

those maturities may not coincide with liability amounts. The recommended hybrid preserves essential dated payments while allowing residual exposures to be managed within approved risk limits.

Liquidity deserves separate attention because rebalancing itself can add cost. AOFM's historical analysis links Treasury Bond liquidity to issuance concentration, market participation and active trading around futures-related maturities (AOFM 2020). Those structural observations support checking executable depth, but the 2020 market sizes and spreads should not be presented as current estimates. For the group portfolio, a marginally cheaper initial purchase may be unattractive if later adjustment requires a wide spread or cannot be executed in the needed volume.

2.5 Execution outcome and practical learning

Actual execution cannot be evaluated from the briefing and sample report. The required evidence includes purchased-asset screenshots, dated LSEG prices and the workbook's Liability and Asset Chart. Without these records, reporting complete liability coverage, a realised saving or particular trading lessons would be unsupported. The outstanding assessment is to reconcile intended quantities with purchases, identify any execution deviations, and explain their effects on cost and payment coverage. Reflection must come from the group's genuine experience and contribution records.

3. Part 2: Australian Government Bond Yield-Curve Scenarios

3.1 Observed baseline and forecast method

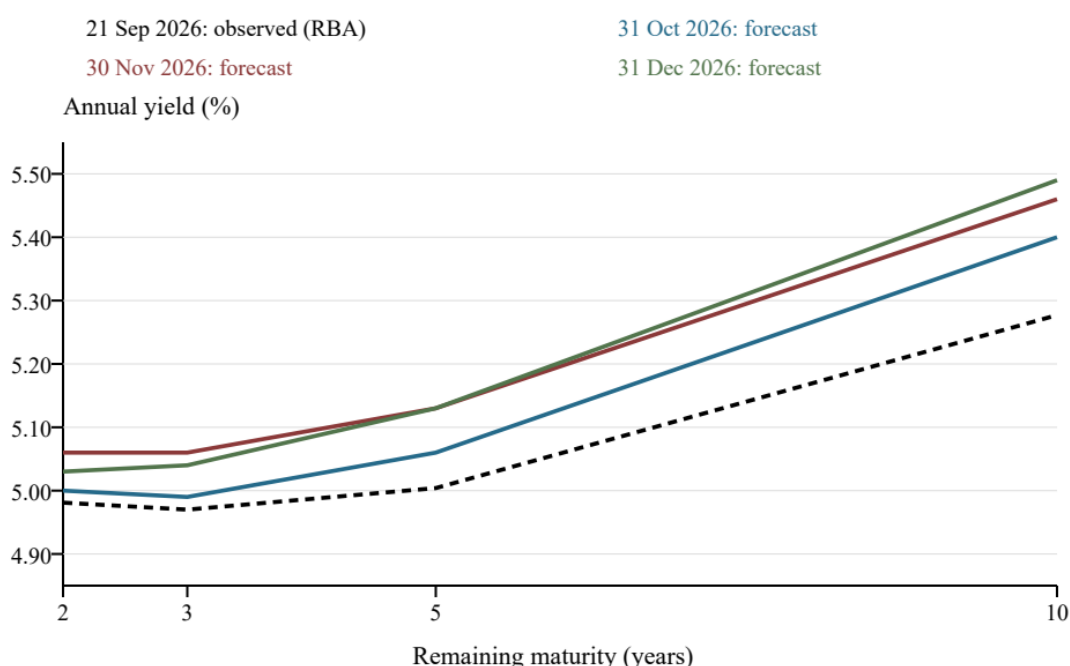
RBA Table F2 provides interpolated Australian Government Bond yields at two, three, five and ten years. The 21 September observations are 4.981%, 4.970%, 5.004% and 5.277%, respectively (RBA 2026b). The curve is almost flat across the shorter maturities and rises towards ten years; its ten-minus-two-year spread is 29.6 basis points. These are government bond yield observations, not a bootstrapped zero-coupon curve. They should not be used interchangeably with spot discount rates. The final assessment still requires the prescribed LSEG comparison.

Table 1: Observed government bond yields and central forecasts (%)

Maturity	21 Sep observed	31 Oct forecast	30 Nov forecast	31 Dec forecast
2 years	4.981	5.00	5.06	5.03
3 years	4.970	4.99	5.06	5.04
5 years	5.004	5.06	5.13	5.13
10 years	5.277	5.40	5.46	5.49

Source: RBA (2026b), Table F2, for observations; original research scenarios for forecasts. RBA data substitute for LSEG pending verification.

The latest observation retrieved in the downloaded file is 30 September: 4.942%, 4.922%, 4.983% and 5.344%. Relative to 21 September, shorter yields fell while the ten-year rose 6.7 basis points. Consequently, the model does not mechanically add the September cash-rate increase to every baseline yield. It starts from the latest retrieved curve, then applies modest tenor-specific changes reflecting the assumptions below. Straight lines connect observed maturities for presentation; they do not imply additional measured tenors.

Figure 1: Observed baseline and three forecast yield curves

Source: RBA (2026b) and original research scenarios. The chart covers four comparable maturities; forecast values are judgements, not market observations.

3.2 Macroeconomic evidence and assumptions

Monetary policy remains restrictive. On 29 September, the RBA increased the cash rate by 25 basis points to 4.60%, citing persistent inflation pressures and retaining scope for further

tightening (RBA 2026e). The central scenario assumes a 4.60% October rate, an increase to 4.85% in November and no December change. These are the report's assumptions rather than an RBA commitment. With no scheduled October decision and meetings on 2–3 November and 7–8 December, the scenario timing follows the published calendar (RBA 2026a).

The ABS August release, published on 30 September, reports annual headline inflation of 4.0% and trimmed-mean inflation of 3.6%. Headline inflation increased from 3.5% in July, while the underlying measure was unchanged (ABS 2026b). The combination argues against assuming rapid disinflation. The central case allows energy-related pressure to remain relevant without assuming a permanent acceleration in every CPI component. Persistent underlying inflation supports restrictive short-rate expectations, while uncertainty about how long inflation lasts justifies a modest increase in longer-term compensation.

Labour conditions give a more mixed signal. August employment rose by 39,500, participation reached 67.1%, and unemployment was 4.6% (ABS 2026c). Higher unemployment alongside employment growth is not equivalent to an outright contraction in jobs: changing labour supply matters. The reported monthly unemployment movement uses unrounded estimates, so it need not equal subtraction of the displayed rounded rates. The forecast assumes gradual easing in labour demand, limiting the number of additional rate rises rather than requiring an immediate cut. One monthly release is insufficient to establish a persistent trend.

June-quarter real GDP grew 0.4% and 2.1% through the year, while household saving was 6.5% (ABS 2026a). This indicates continued expansion with limited quarterly momentum. The RBA's August outlook projected December annual GDP growth of 1.4% and trimmed-mean inflation of 3.3%, under assumptions finalised on 5 August (RBA 2026d). These projections provide context rather than immutable targets. Subsequent inflation information warrants more caution about the near-term policy path, while slow growth argues against projecting repeated large increases indefinitely.

Exchange rates and overseas rates affect the balance. The August financial-conditions assessment reported a 1.2% trade-weighted depreciation since May, while the currency remained approximately 5% above its start-year level (RBA 2026c). The central case assumes no further sustained depreciation; renewed weakness is an upside inflation risk. International yield increases can also support Australian long yields through relative-return demands. A stable currency cannot remove all imported price pressure, and a weaker currency need not mechanically imply a specified cash-rate increase.

3.3 Curve shapes, calibration and uncertainty

By 31 October, the scenario is 5.00%, 4.99%, 5.06% and 5.40% across the four maturities. It retains a shallow trough at three years and an upward slope thereafter. Relative to 30 September, the increases are approximately 5.8, 6.8, 7.7 and 5.6 basis points. The calibration assumes that persistent inflation prevents an early rally, while existing high short yields already incorporate substantial restrictive-policy expectations. The forecast is therefore more restrained than a uniform 25-basis-point shift.

By 30 November, yields reach 5.06%, 5.06%, 5.13% and 5.46%. The short segment is flat and the curve rises beyond three years. The assumed November tightening supports the two- to five-year area, while inflation and overseas-rate uncertainty keep the long end elevated. By 31 December, the corresponding yields are 5.03%, 5.04%, 5.13% and 5.49%. Modestly lower short yields reflect a policy pause and weaker growth expectations, but the ten-year remains higher. The ten-minus-two-year spread widens to 46 basis points: moderate steepening relative to September.

This is a judgemental scenario, not an econometric estimate or a forecast with measured probabilities. Two stresses test its dependence on assumptions. Persistent inflation assumes the December cash rate reaches 5.10%, with December yields 25, 25, 20 and 20 basis points above the central case. A growth shock assumes easing becomes plausible, with yields 40, 40, 30 and 25 basis points below central values. These shifts are transparent sensitivity choices. They describe alternative management conditions rather than statistical confidence intervals, and must be reassessed when new information arrives.

3.4 Portfolio implications

The central case favours preserving liquidity and avoiding additional unhedged long-duration exposure in discretionary assets. It does not justify selling bonds needed to fund matched obligations. Modified duration and convexity approximate price sensitivity, but aggregate duration alone assumes a common yield movement and can conceal curve-specific risks (CFA Institute 2026d). For a purely illustrative bond with modified duration 8 and convexity 75, a 25-basis-point rise implies a price change of approximately -1.98% , excluding income and trading costs. Actual portfolio effects require measured instrument sensitivities and liability values.

$$\Delta P / P \approx -D_{\text{mod}} \times \Delta y + \frac{1}{2} \times \text{Convexity} \times (\Delta y)^2 = -8 \times 0.0025 + \frac{1}{2} \times 75 \times 0.0025^2 = -1.98\%$$

4. Recommendations

First, reconcile the liability workbook with executed holdings before approving any statement of minimum cost or complete funding. Apply actual settlement terms and trade-size restrictions, then compare feasible alternatives. Second, ring-fence essential dated cashflows and apply the hybrid approach only to residual flexibility permitted by the mandate. Third, retain a cautious duration position in discretionary assets while inflation persists; reassess after the October CPI release and November policy meeting. Increase duration only when evidence supports easing and funding remains protected. Finally, monitor short- and long-maturity sensitivities separately: a steepening forecast cannot be managed reliably using a single portfolio-duration number. Recommendations should be implemented against documented limits, not forecast certainty.

5. Conclusion

Cashflow matching offers a verifiable funding discipline, but its success depends on executable prices, payment timing and feasible holdings. The supplied materials permit a researched construction analysis, not a conclusion about the group's actual trading outcome. The macroeconomic evidence supports a restrictive near-term policy scenario and modestly higher long yields through December, with growth weakness limiting short-rate escalation. Protecting liability coverage while managing discretionary duration is the preferred response. Final assessment completion requires the actual workshop workbook, genuine execution reflection and LSEG evidence to replace the disclosed research substitutions.

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Appendices

Appendix A: Illustrative cashflow calculation

All bonds, prices and liabilities below are hypothetical and expressed in generic currency units. This example demonstrates quantity selection only. It is not Week 11 evidence, does not use LSEG prices and cannot replace the required trading workbook. Three annual liabilities of 100 are funded by annual-coupon bonds with 100 face value and a coupon of 4. Bond A matures in year 1, B in year 2 and C in year 3; assumed full purchase prices are 98, 97 and 96. Fractional quantities, no fees, no default and zero surplus reinvestment are assumed.

$$x_C = 100 / 104 = 0.961538$$

$$x_B = (100 - 4x_C) / 104 = 0.924556$$

$$x_A = (100 - 4x_B - 4x_C) / 104 = 0.888996$$

$$\text{Initial cost} = 98x_A + 97x_B + 96x_C = 269.11$$

Year	Bond A inflow	Bond B inflow	Bond C inflow	Liability / total inflow
1	92.4556	3.6982	3.8462	100.0000 / 100.0000
2	0.0000	96.1538	3.8462	100.0000 / 100.0000
3	0.0000	0.0000	100.0000	100.0000 / 100.0000

The quantities solve the exact-match equations for this restricted three-bond illustration. They are not a demonstrated global minimum over an actual tradable universe. Minimum-cost claims require the optimisation and feasibility checks described in section 2.3. Rounding these quantities without recalculation may create a shortfall.

Appendix B: Forecast calibration and stress values

Four tenors are taken directly from RBA F2. Forecasts are original scenario assumptions anchored to the retrieved 30 September curve. Forecasts use annual percentage yields; 1 basis point = 0.01 percentage point. No probability is assigned to any scenario.

Tenor	30 Sep observed	Dec central	Dec stress inflation	Dec growth stress
2 years	4.942	5.03	5.28	4.63
3 years	4.922	5.04	5.29	4.64
5 years	4.983	5.13	5.33	4.83
10 years	5.344	5.49	5.69	5.24

RBA F2 notes describe these observations as the RBA's assessments of closing bond yields, intended as a convenience for academic research. They may differ from market prices and may be revised. The report therefore does not treat them as executable quotations. The retrieved file contained selected dates through 30 September; it was not extended by inventing October observations.

Appendix C: Release dates and evidence status

Evidence	Date / period	Status
RBA baseline yields	21 September 2026	Retrieved from Table F2; LSEG replacement pending
RBA latest retrieved curve	30 September 2026	Retrieved from Table F2
ABS GDP	June quarter; released 2 September	Verified official release
ABS Labour Force	August; released 24 September	Verified official release
RBA cash-rate decision	29 September	Verified official release
ABS CPI	August; released 30 September	Verified official release
Group trades and liabilities	Week 11	Not supplied; cannot verify outcomes
LSEG bond prices and curve	Required by briefing	Not supplied
Workbook Liability and Asset Chart	Required by briefing	Not supplied
Group reflection and contributions	Actual member experience	Not supplied; members must confirm

Appendix D: Academic integrity and completion note

This report was prepared with AI assistance for research synthesis, drafting, forecast scenarios, calculations and formatting. It is a research draft, not a record of work already performed by the group. Students must check the course-specific permission for these uses, verify and understand the analysis, supply their genuine group evidence and acknowledge retained AI outputs as required by RMIT. The sample student report was used for visual layout only; its wording, financial data, student details and references were not transferred.

RMIT guidance: [Using AI in learning and assessments](#)

Before submission, incorporate actual trades and liabilities into Part 1, replace the illustrative calculation with the group's reconciled outcomes, include required screenshots and the workbook chart, verify the LSEG baseline and revise forecasts if its values differ. Add genuine learning reflections, group details and an agreed contribution statement. Confirm the word-count treatment of tables and appendices, update the contents and remove this completion note only after those steps are resolved.

Research access note: the 1990 actuarial paper is cited for its publisher-indexed description of the linear-programming approach. Direct PDF retrieval was unavailable during this run; the full article should be obtained through the library before relying on further claims from it. All macroeconomic figures were checked against official releases and the RBA yield workbook was downloaded directly.